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SES, Product and Innovation Engineering

Dear Hiring Team,

I am applying for the AI Engineer Intern role on the Product and Innovation Engineering team. I am a Master's student in Computer Science at UMass Amherst who has spent the last year building exactly what this role is about: LLM agents that use tools, RAG pipelines grounded on real knowledge bases, and full-stack AI applications that have to work, not just demo. Your emphasis on production-ready systems over proof-of-concept is the reason I am excited about this team.

Most of my recent work is agent and retrieval engineering. I built AgenticSearch, a multi-stage LLM agent that plans typed retrieval, calls search and extraction tools, verifies its own outputs, and returns cell-level provenance (value, source, and confidence) for every result, so a reviewer can trust where each answer came from. I hardened the FastAPI service with 150+ automated tests and provider routing/fallback across OpenAI and Groq as a reliability guardrail, with deterministic extraction before any LLM fallback. That habit of grounding agent behavior in tests, guardrails, and provenance is what I would bring to building agents against your proprietary knowledge bases and custom APIs.

On the RAG side, I led the multi-view retrieval component of a knowledge-grounded pipeline for multi-hop QA: query-side sub-question decomposition, per-view embedding retrieval, Reciprocal Rank Fusion, cross-encoder reranking, and MMR, built on llama-index and sentence-transformer embeddings. It lifted supporting-fact recall +2.68% over a strong baseline on a 4,905-question evaluation with proper confidence intervals. I have also built a conversational research agent that uses tool calls to fetch sources, streams responses to the client over WebSocket in real time, and keeps persistent agent memory and durable multi-step workflows so long-running tasks survive restarts and tool failures, the streaming and memory patterns your description calls out.

I am comfortable across the full stack this role spans: Python and FastAPI backends, React frontends, and REST APIs (I shipped 20+ Node.js REST APIs and integrated LLMs into a live customer-facing chat product during an earlier internship, and an ML-data platform used daily by an AI team at Wysa). I work in Agile teams and recently took part in a measured GitHub Copilot productivity pilot, so AI-assisted development and sprint ceremonies are second nature.

I want to be straight about two gaps relative to your nice-to-haves: my retrieval work has used embeddings with FAISS and llama-index rather than a managed vector database like Pinecone or AWS OpenSearch, and I have used AWS services (S3, KMS, SQS) but not Bedrock yet. Given how quickly I picked up the agent, RAG, and streaming stack on my own, I am confident I can close those in an AWS-first environment fast. I would welcome the chance to contribute to the team's AI products.

Sincerely,
Deva Anand